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Heme's relevance genuine? Re-visiting the roles of TANGO2 homologs including HRG-9 and HRG-10 in *C. elegans*

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eLife Assessment

This **valuable** study provides **solid** evidence that supports TANGO2 homologs, including HRG-9 and HRG-10, can play a role in cellular bioenergetics and oxidative stress homeostasis. It also challenges the previously reported role of TANGO in heme transport and paves the way for future mechanistic studies addressing the mechanisms of how TANGO2 regulates oxidative stress homeostasis. The strengths include the use of different model systems, genetic tools, behavioral assays and efforts by the authors in using the same reagents to reproduce results of other groups.

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Abstract

Mutations in the *TANGO2* gene are associated with a severe neurometabolic disorder in humans, often presenting with life-threatening metabolic crisis. However, the function of TANGO2 protein remains unknown. It has recently been proposed that TANGO2 transports heme within and between cells, from areas with high heme concentrations to those with lower concentrations. Here, we demonstrate that prior heme-related observations in *Caenorhabditis elegans* lacking TANGO2 homologs HRG-9 and HRG-10 may be better explained by a previously unreported metabolic phenotype, characterized by reduced feeding, decreased lifespan and brood sizes, and poor motility. We also show that several genes not implicated in heme transport are upregulated in the low heme state and conversely demonstrate that *hrg-9* in particular is highly responsive to oxidative stress, independent of heme status. Collectively, these data implicate bioenergetic failure and oxidative stress as potential factors in the pathophysiology of TANGO2 deficiency, in alignment with observations from human patients. Our group performed several experiments in yeast and zebrafish deficient in TANGO2 homologs and was unable to replicate prior findings from these models. Overall, we believe there is insufficient evidence to support heme transport as the primary function for TANGO2.

Impact statement

Cross-species examination of TANGO2 homologs demonstrates that phenotypes previously attributed to altered heme trafficking may instead reflect broader disturbances in mitochondrial function and cellular homeostasis.

Introduction

Transport and Golgi Organization 2 (TANGO2) Deficiency Disorder (TDD) is a severe, progressive, neurodegenerative disease primarily affecting children and young adults, frequently accompanied by recurrent, life-threatening metabolic crises.^{1–4} While TANGO2 is conserved in species across all three domains of life, its cellular function has not been fully elucidated. Initially identified in a screen of *Drosophila* genes which, when depleted, precipitate ER-Golgi fusion defects,⁵ TANGO2 has since been shown to localize to both the cytoplasm and the mitochondria.^{6,7} Mutations in TANGO2 have been associated with altered mitochondrial respiration and morphology,⁸ dysregulated lipid homeostasis,^{9,10} and defects in fatty acid oxidation⁸ in multiple models. Treatment with supplemental pantothenic acid, the precursor of coenzyme A (CoA), has been shown to improve lipid profile abnormalities in cells,¹⁰ rescue phenotypic defects in a *Drosophila* TDD model,¹¹ and potentially reduce metabolic crises in patients,¹² implicating reduced CoA, lipid dysregulation, and impaired mitochondrial function as putative drivers of TDD symptoms.

A recent study by Sun et al. suggested that two *Caenorhabditis elegans* homologs of TANGO2, HRG-9 and HRG-10 (Heme-Responsive Gene), play a critical role in heme export from intestinal cells and lysosome-related organelles and that TANGO2 exhibits a similar function in mobilizing heme from mitochondria in other eukaryotic organisms including yeast.¹³ Two subsequent studies also reported that TANGO2 is involved in heme transport, albeit via discordant mechanisms.^{14,15} The notion that defective heme transport underlies the pathophysiology of TANGO2 deficiency marks a sizeable shift from the emerging scientific consensus that TANGO2 is critical for CoA and lipid homeostasis. Therefore, our primary objective was to validate key findings pertaining to heme transport in *C. elegans* while also replicating key findings in zebrafish and yeast lacking TANGO2 homologs. We hypothesized that, if present, defects in heme transport may be downstream features of aberrant cellular metabolism and thus may not be central to the pathophysiology of TDD.

Results

***C. elegans* lacking TANGO2 homologs (double knockout; DKO) demonstrate modest survival benefit upon toxic heme analog exposure**

We first sought to validate results from prior experiments using heme analogs in worms lacking both HRG-9 and HRG-10 (double knockout; DKO). In previous studies, DKO worms exposed to 1 μ M concentration of the toxic heme analog gallium protoporphyrin IX (GaPP) showed a significant survival benefit in contrast with nearly uniform lethality in wild type (Bristol N2) worms.¹³ We exposed gravid DKO and N2 nematodes to GaPP at varying concentrations, removing P₀ worms at 24 hours, and counting alive and dead F₁ progeny at 72 hours. While we required relatively higher concentrations of GaPP to achieve lethality, all strains exhibited a clear dose-dependent reduction in survival (Figure 1A [↗](#)). A statistically significant group effect was observed at 2 μ M, with DKO worms showing a modest relative survival benefit compared to N2.

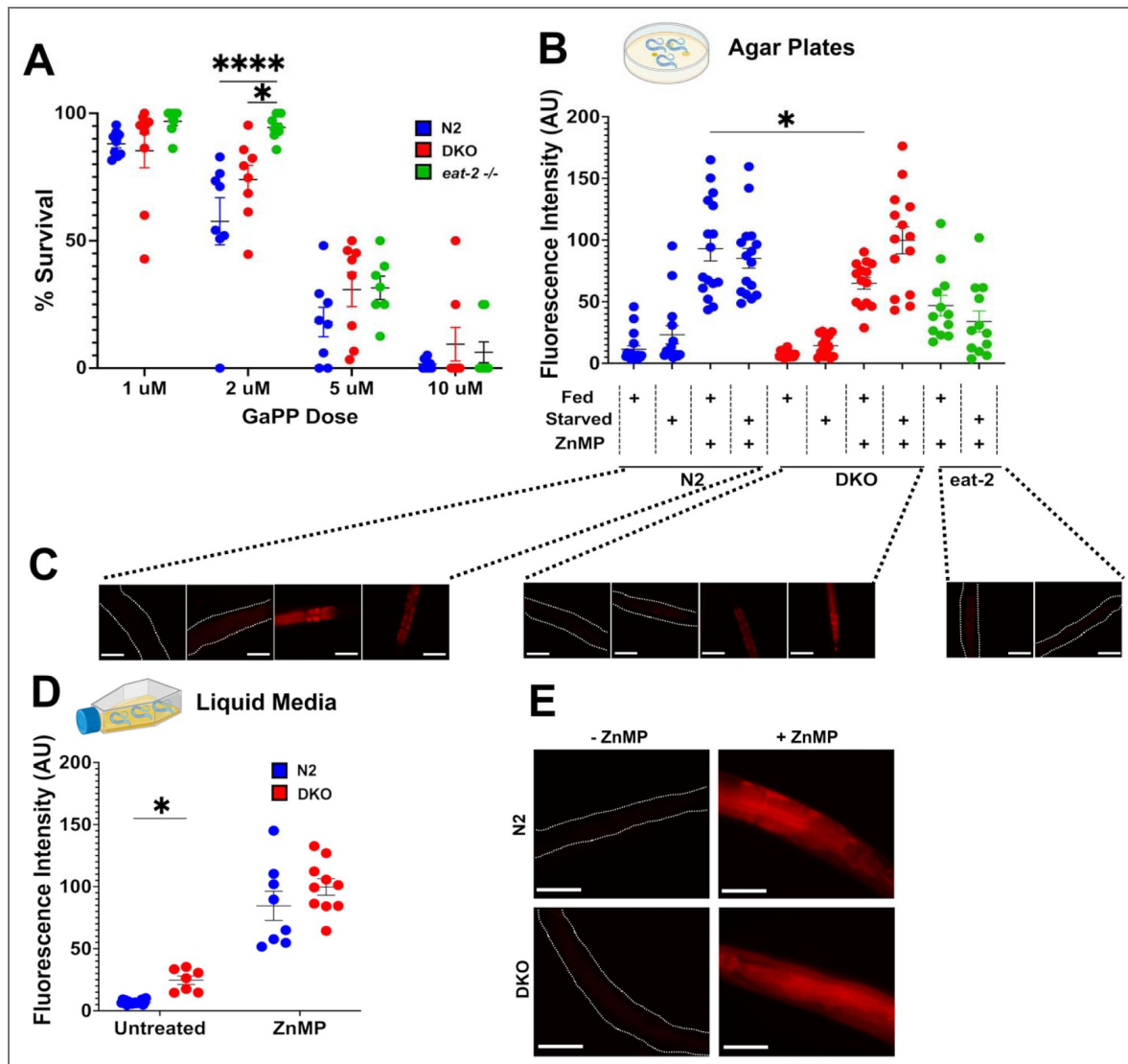


Figure 1. Reduced ingestion of toxic and fluorescent heme analogs may explain differences observed between wildtype *C. elegans* and worms lacking HRG-9 and HRG-10 (double knockout; DKO).

A) Survival of N2, DKO, or *eat-2* knockout worms exposed to 1, 2, 5, or 10 μ M GaPP for 72 hours. Each dot represents the number of offspring laid by one adult worm on one GaPP-treated plate after 24 hours. N=3 independent experiments. * $p < 0.05$, **** $p < 0.0001$ **B)** Quantification of fluorescent staining in N2, DKO, and *eat-2* worms grown on OP50 *E. coli* plates under fed and starved conditions with or without 40 μ M ZnMP treatment. A.U. arbitrary units, N=15 worms analyzed over 3 independent experiments. * $p < 0.05$. **C)** Representative images of red fluorescence in N2, DKO, and *eat-2* worms grown on OP50 *E. coli* plates. Scale bars represent 50 μ M. **D)** Quantification of fluorescent staining in N2 and DKO cultured in normal heme axenic media with or without 40 μ M ZnMP treatment. A.U. arbitrary units, N=7-15 worms analyzed over 3 independent experiments. * $p < 0.05$. **E)** Representative images of red fluorescence in N2 and DKO worms grown in liquid media. Scale bars represent 50 μ M.

DKO *C. elegans* demonstrate increased lawn avoidance and reduced pharyngeal pumping

In maintaining the DKO strain under basal conditions (i.e., no GaPP exposure), we incidentally observed relatively intact *E. coli* lawns on plates housing DKO worms compared to N2 plates. On closer examination, DKO worms demonstrated several previously undescribed phenotypic features including lawn avoidance, a greater propensity for crawling off plates, reduced pharyngeal pumping, and decreased survival (Figure 2A, B, and F). Based on these observations, we hypothesized that the GaPP survival difference in our initial experiment might have been driven, at least in part, by reduced GaPP consumption. We therefore also examined the effect of GaPP exposure on worms lacking a pharyngeal acetylcholine receptor subunit (*eat-2(ad465)*). The *eat-2* strain exhibits reduced pharyngeal pumping and has been used extensively as a model of dietary restriction.^{16,17} Of the three strains tested, survival was highest in *eat-2* mutants (Figure 1A) despite the fact that these worms have no known defects in heme metabolism or transport.

DKO *C. elegans* demonstrate reduced ZnMP fluorescence

DKO nematodes at the L4 larval stage were previously shown to accumulate the fluorescent heme analog zinc mesoporphyrin IX (ZnMP) in intestinal cells in low-heme (4 μ M) liquid media. While attempting to replicate this experiment, we observed that both wildtype and DKO nematodes entered L1 arrest under these conditions. Therefore, to allow for developmental progression, we grew worms on standard OP50 *E. coli* plates and in media containing physiological levels of heme (20 μ M). We then examined whether differences in ZnMP uptake persisted under these basal conditions. DKO worms grown on ZnMP-treated *E. coli* plates displayed significantly reduced intestinal ZnMP fluorescence compared to N2 (Figure 1B and C). Using basal heme media with ZnMP, there was no significant difference in ZnMP fluorescence between DKO and wildtype nematodes, although DKO worms grown in media without ZnMP exhibited significantly higher autofluorescence (Figure 1D and E). To test whether autofluorescence may have contributed to the higher fluorescent intensities previously reported in heme-deficient DKO worms, we repeated this experiment on agar plates under starved conditions but did not observe a difference between groups (Figure 1B).

DKO *C. elegans* demonstrate multiple features suggestive of bioenergetic dysfunction

In assessing brood survival in the GaPP assay, we also observed significantly smaller starting broods for DKO nematodes, a previously unreported finding that persisted in the absence of GaPP (Figure 2C). As oogenesis and egg-laying require a high energetic expenditure for gravid *C. elegans*, reduced brood size is a known feature of several metabolically impaired nematode strains.^{18,19} We also observed reduced movement on the plate from DKO worms and decided to further characterize the worms' capacity for movement by subjecting them to a swim exhaustion assay. Nematodes were placed in isotonic M9 buffer and scored on their swimming ability at 4 minute intervals. While *C. elegans* are typically able to swim continuously for up to 90 minutes in M9 media,²⁰ we observed that the DKO worms quickly became exhausted and could not maintain normal swimming shortly after being placed in media (Figure 2D). We further quantified the rate of thrashing in M9 media during the first 4-minute interval and found that DKO worms thrashed 61% slower compared to N2 (Figure 2E).

Oxidative stress is a driver of TANGO2 homolog transcription

Given previous observations implicating oxidative stress as a hallmark feature of TDD,⁸ we next sought to examine what other genes were enriched under low heme conditions. We reanalyzed the RNA-seq dataset generated by Sun et al., employing the Empirical Analysis of Gene Expression in R (edgeR) package on raw counts to accurately perform between-group comparisons across low (2 μ M), optimal (20 μ M), and high (400 μ M) heme conditions. We extracted the top 500 enriched genes and plotted those that showed significantly increased expression in the low heme state,

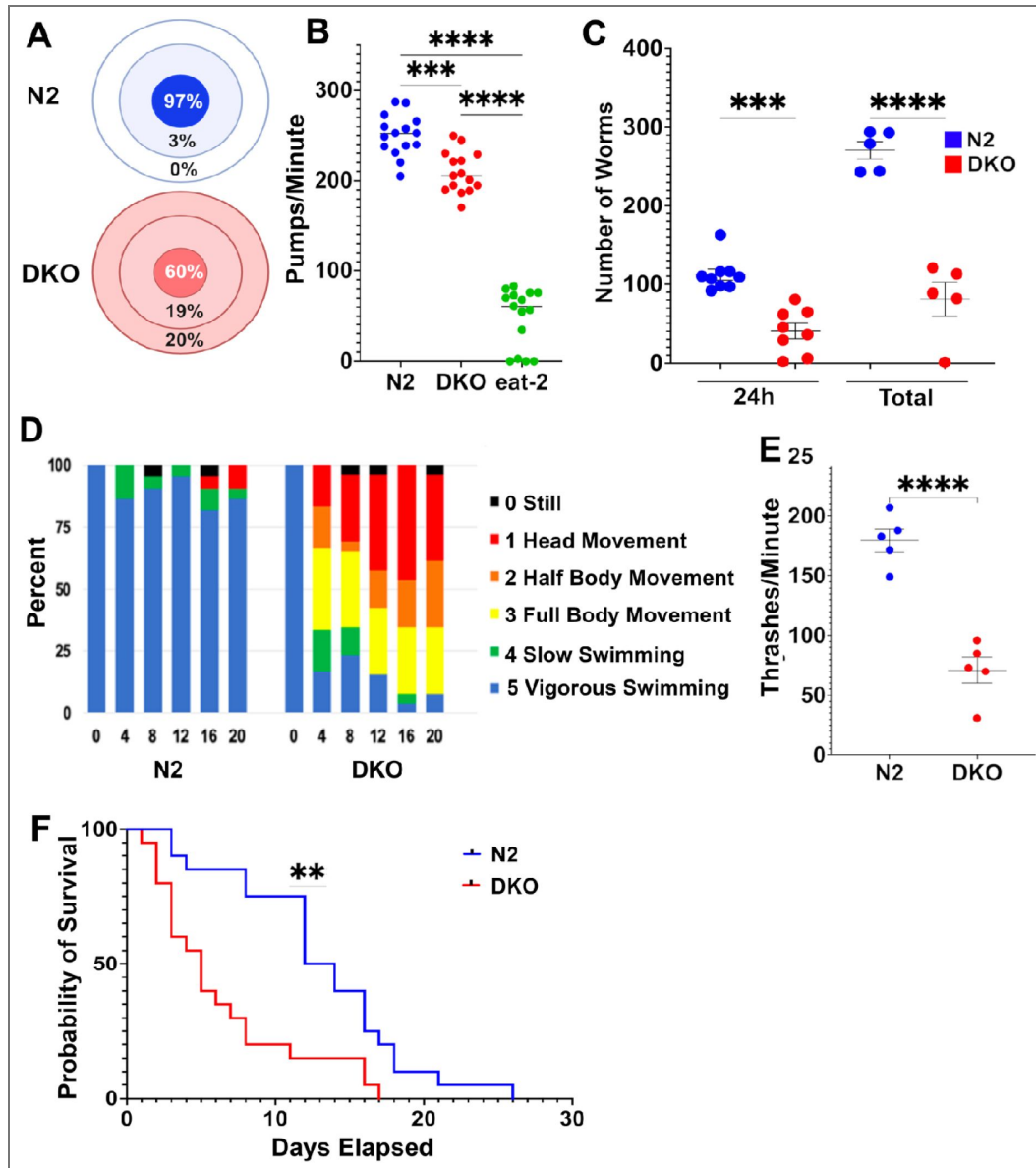


Figure 2. DKO nematodes demonstrate lawn avoidance and reduced pharyngeal pumping, brood sizes, motility, and survival.

A) Proportion of N2 or DKO worms present on OP50 lawn (innermost ring), off OP50 lawn (middle ring), or missing or dead from NGM plate (outer ring). N=180 worms over 3 independent experiments **B**) Number of pharyngeal pumps in a one-minute period in N2, DKO, and *eat-2* knockout worms. N=15 worms over 3 independent experiments **C**) Number of viable offspring laid by single adult N2 or DKO worms either after 24 hours of egg lay or across total egg-laying period (5 days). N=8 broods for 24-hour counts, N=5 broods for total brood size counts. **D**) Swimming behavior of N2 and DKO worms over a 20 minute interval. Worms were observed at 4 minute intervals and scored from 0-5 on swimming intensity. Bars represent the proportion of worms at each score. **E**) Quantification of thrashes after 4 minutes in M9 buffer. **F**) Longevity of N2 and DKO *C. elegans* observed from L4 larval stage. N=30 worms over 3 independent experiments. (** p<0.01, *** p<0.001, **** p<0.0001).

based on computational clustering (N=134; Figure 3A). Several genes with no known heme-related functions demonstrated stronger relative expression and higher likelihood ratios of conditional effect than did *hrg-9*. Furthermore, gene ontology analysis of genes with similar enrichment to *hrg-9* revealed a wide spectrum of biological processes and cellular roles, including but not limited to collagen deposition, cellular detoxification, and lipid binding (Figure 3B). To test what alternate forms of cell stress might induce *hrg-9* and *hrg-10* expression, we exposed N2 nematodes to heat (34°C), starvation (24 hours without OP50), and paraquat (25 mM), a potent generator of superoxide, with subsequent measurement of transcriptional enrichment by way of RT-qPCR. We observed a 12-fold enrichment of *hrg-9* after paraquat exposure, suggesting that its expression may be linked to oxidative stress more broadly and is not uniquely driven by heme levels (Figure 3C).

Yeast deficient in TANGO2 homolog YGR127w exhibit normal growth

As Sun et al. also examined the function of TANGO2 homologs in yeast (YGR127w) and zebrafish (*tango2*), we sought to validate their findings in these models. It was previously reported that the *ygr127w* yeast knockout exhibits a severe temperature sensitive growth defect and impaired heme distribution. We also observed a growth defect in the strain generated by Sun et al.; however, two separate strains on different backgrounds (BY4741 and BY4742) and a third strain generated by our lab (W303-1a) showed normal growth (Figure 4A). The growth defect in the initial yeast strain reported by Sun et al. was also not rescued with YGR127w complementation. As the background for this strain is known to be prone to mitochondrial genome instability,²¹ we hypothesize that this line may harbor a secondary mutation.

Myofiber defects in *tango2*-deficient zebrafish

In zebrafish, Sun et al. reported no discernable phenotype in *tango2*^{-/-} fish bred from heterozygous parents but observed severe skeletal muscle damage in *tango2*^{-/-} larvae from *tango2*^{-/-} parents. In a recent study, we showed increased lethality and reduced phospholipid and triglyceride levels in *tango2*^{-/-} fish obtained from heterozygous parents.²² *tango2*^{-/-} larvae from two different alleles (*tango2*^{bwh210} and *tango2*^{bwh211}) exhibited defects in myofiber organization irrespective of heterozygous or homozygous parents (Figure 4B) but lacked the striking muscle damage previously reported.

Discussion

Our data, coupled with clinical observations in TDD, do not support a primary role for TANGO2 as a heme chaperone. In *C. elegans* lacking TANGO2 homologs, we were unable to fully replicate prior findings of defective heme transport using toxic and fluorescent heme analogs. Parallel studies in yeast and zebrafish failed to reproduce previously reported phenotypes in growth and muscle fiber integrity, respectively.

Exposing worms to GaPP, a toxic heme analog, we observed that nematodes deficient in HRG-9 and HRG-10 displayed increased survival compared to WT worms, consistent with prior work,¹³ though the between-group difference was markedly smaller in our study. We required higher GaPP concentrations to induce lethality, potentially due to product vendor differences, but did observe a clear dose-dependent effect across strains. Although it was previously proposed that the survival benefit seen in worms lacking HRG-9 and HRG-10 resulted from reduced transfer from intestinal cells after GaPP ingestion, our data suggest the reduced lethality is more likely due to decreased environmental GaPP uptake. Supporting this notion, DKO worms exhibited lawn avoidance, reduced pharyngeal pumping, and modestly lower intestinal ZnMP accumulation when exposed to this fluorescent heme analog on agar plates. In liquid media, DKO worms demonstrated higher fluorescence, but only in ZnMP-free conditions, suggesting the presence of gut granule autofluorescence. Furthermore, survival following exposure to GaPP was highest in *eat-2* mutants, despite heme trafficking being unaffected in this strain.

Figure 3. RNA-seq and qPCR analysis show that *hrg-9* and *hrg-10* are not uniquely heme responsive but instead may be preferentially induced under conditions of oxidative stress.

A) Analysis of top 500 genes with differential expression under low heme (2 μ M) and high heme (400 μ M) conditions. Outline represents 134 relevant genes identified by cluster analysis. R186.1 is the alternative sequence name for *hrg-9*. **B)** Gene ontology analysis identified a variety of biological roles for genes within this cluster **C)** RT-qPCR of *hrg-9* and *hrg-10* under non-heme stress conditions: 24-hour starvation, 4-hour exposure to 34° C heat, and 25 mM paraquat (pqt). N=3 independent experiments.

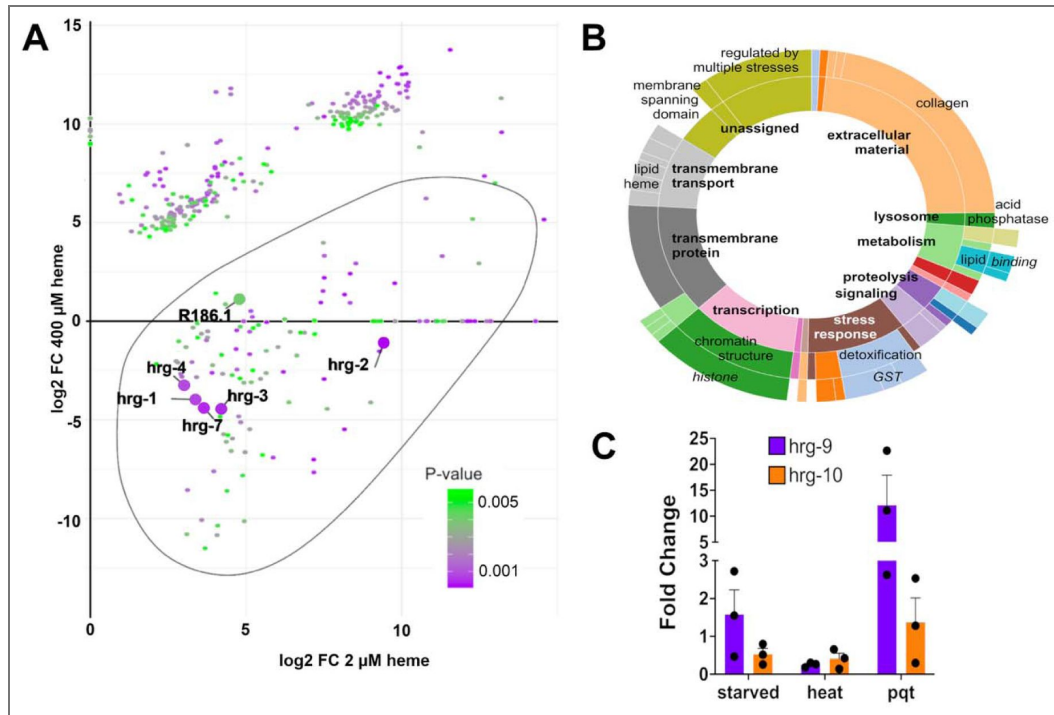
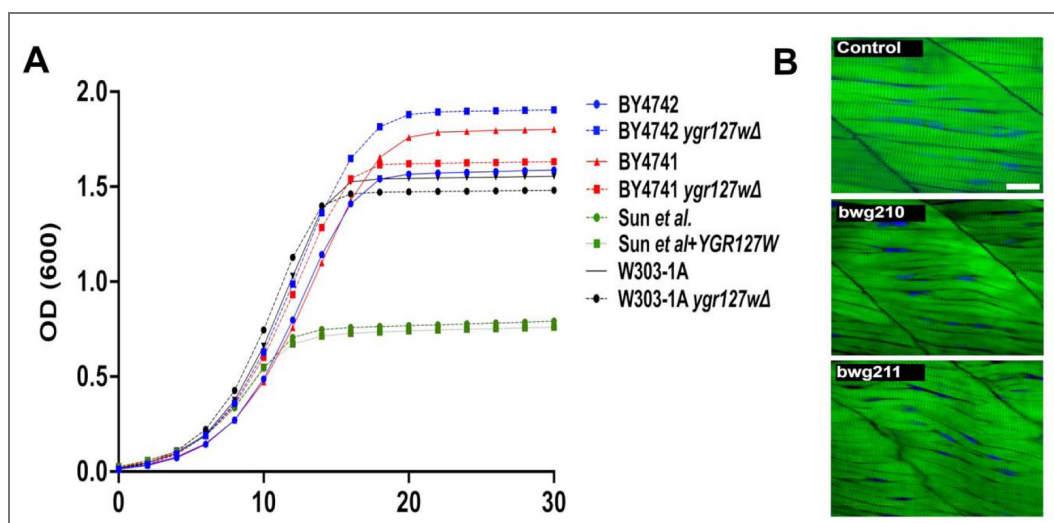


Figure 4. Growth and muscle fiber integrity in yeast and zebrafish models of TANGO2 deficiency fail to replicate previously reported phenotypes.

A) Yeast growth curves. Different strains were grown in SC medium or SC medium lacking histidine at 25° C. **B)** Whole mount phalloidin staining of control and two strains of *tango2*^{-/-} zebrafish (*bwg*²¹⁰ and *bwg*²¹¹). Myofibers in mutants lack the parallel organization observed in controls but do not demonstrate significant myofiber breakdown or damage. Representative images; N=8-10 in each group. Scale bar=5 mm.



In addition to altered pharyngeal pumping, DKO worms displayed multiple previously unreported phenotypic features, suggesting a broader metabolic impairment and reminiscent of some clinical manifestations observed in patients with TDD. Elucidating the mechanisms underlying this phenotype, and whether they reflect a core bioenergetic defect, is an active area of investigation in our lab. Several *C. elegans* heme-responsive genes have been characterized, revealing relatively specific defects in heme uptake or utilization rather than broad organismal dysfunction. For example, *hrg-1* and *hrg-4* mutants exhibit impaired growth only under heme-limited conditions,²³ and *hrg-3* loss affects brood size and embryonic viability specifically when maternal heme is scarce.²⁴ By contrast, *hrg-9* and *hrg-10* mutants exhibit the most severe phenotypes of the *hrg* family, to date, including reduced pharyngeal pumping, decreased motility, shortened lifespan, and smaller broods, even when fed a heme-replete diet.

Laboratory abnormalities in human patients TDD include abnormal acylcarnitine profiles, hyperammonemia, and elevated creatinine kinase levels during metabolic crises,^{3,12} while abnormalities associated with defective heme transport (e.g., erythrocyte membrane defects, low hemoglobin levels) have not been observed in patients with this condition. Strikingly, retrospective data suggest that patients with TDD receiving B-vitamin supplementation inclusive of pantothenic acid, a precursor of CoA, do not experience metabolic crises¹² and show substantial improvement in other domains as well. Pantothenic acid supplementation also yielded full phenotypic rescue in a *Drosophila* model of TDD.¹¹ It is difficult to reconcile how this treatment would be beneficial in a condition characterized by dysregulated heme trafficking.

Heme is a hydrophobic molecule; thus, it is plausible that if TANGO2 and its homologs are involved in lipid binding, as was recently demonstrated in Hep2G cells,⁷ these proteins may also weakly bind heme.¹⁴ Han et al. demonstrated that a bacterial heme homolog, HtpA, directly binds heme and is necessary for cytochrome c function.¹⁴ We would note, however, that TANGO2 was not among the 378 heme-binding proteins identified on a recent proteomic screen of three separate cell lines.²⁵ Jayaram et al. recently proposed that TANGO2 may instead interact with the mitochondrial heme exporter FLVCR1b to release mitochondrial heme without directly binding to heme itself, though this interaction was observed only after heme synthesis was potentiated via d-ALA and iron supplementation, raising questions about the role of TANGO2 under basal heme conditions.¹⁵ Furthermore, the investigators identified GAPDH as the exclusive binding partner of heme upon export through FLVCR1b. How GAPDH, a protein important for multiple cellular functions, including glycolysis, is affected in TDD remains unknown.

In this study, we demonstrated that *hrg-9* expression is strongly induced by paraquat, a generator of superoxide free radical. Prior work has also identified *hrg-9* as a major transcriptional target of the mitochondrial unfolded protein response (mtUPR) through ATFS-1 transcription factor binding.^{26,27} Given that heme is an essential component of cytochromes in the electron transport chain, heme deficiency could plausibly activate *hrg-9* through the induction of mitochondrial stress. Conversely, artificially stimulating heme synthesis may exert a similar transcriptional effect, as excess heme is mitotoxic and can itself trigger a stress response. Indeed, *hrg-9* was also shown to be modestly up-regulated in the high-heme state, unlike other genes in the *hrg* family (Figure 3A [↗](#)), and our reanalysis of an RNA-seq dataset examining transcription under low-heme conditions revealed broad induction of stress-responsive genes with no established role in heme trafficking (Figure 3B [↗](#)).

In summary, our findings challenge the notion that TANGO2 functions as a heme chaperone. Instead, and consistent with growing clinical evidence, they support a model in which TANGO2 may help mitigate cellular stress and maintain mitochondrial function under conditions of redox imbalance (Figure 5 [↗](#)). Clearly, further work is needed to define the precise role of this highly conserved protein as we work to develop effective treatments for patients with TDD.

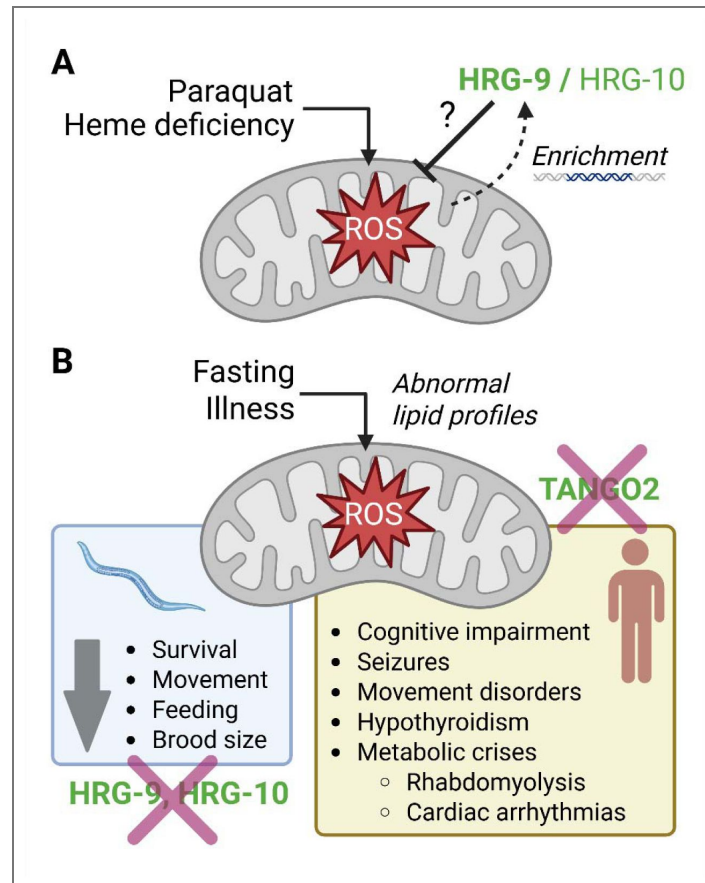


Figure 5. Proposed model of TANGO2 and its homologs acting as stress-responsive mediators of mitochondrial dysfunction.

A) Paraquat exposure and heme deficiency each induce reactive oxygen species (ROS) formation, mitochondrial stress, and enrichment of *hrg-9* in *C. elegans*. TANGO2 and its homologs may help mitigate mitochondrial stress under these conditions, though the exact function of these proteins remains unknown. **B)** Physiological triggers such as fasting or illness also precipitate oxidative stress in the absence of TANGO2. Abnormal lipid profiles have been observed in multiple models of TANGO2 deficiency in the setting of impaired lipid mobilization and reduced fatty acid oxidation. In humans, loss of TANGO2 results in a complex clinical syndrome reminiscent of multiple secondary mitochondrial disorders. In *C. elegans*, loss of *hrg-9* and *hrg-10* induces a phenotype previously observed in nematode strains exhibiting mitochondrial dysfunction. Image created with [BioRender.com/nsnn9pe](https://www.biorender.com/nsnn9pe).

Materials and methods

Worm strains and maintenance

Worm strains used include Bristol N2 (obtained from Caenorhabditis Genetics Center (CGC) at the University of Minnesota), DKO (CCH303 (hrg-9(cck301)V; hrg-10(cck302)V) obtained from C. Chen, and eat-2(ad465), obtained from the Samuelson lab at the University of Rochester. All worms were maintained at 20° C on standard nematode growth medium (NGM) plates with OP50 *E. coli*. All *C. elegans* assays were performed and scored by blinded observers.

GaPP survival assay

Standard OP50 NGM plates treated with 1, 2, 5, or 10 μ M gallium protoporphyrin IX (GaPP; Santa Cruz) after seeding. Plates were swirled to ensure an even distribution of GaPP and allowed to dry completely. Adult worms were placed on plates, permitted to lay eggs for 24 hours, and were then removed. Offspring were assessed 72 hours later for survival and were scored as dead if they did not respond to prodding with a platinum wire worm pick.

Pharyngeal pumping

L4 worms were observed and video recorded for 60 second intervals. The number of pharyngeal pumps per minute was manually counted.

Lawn avoidance

L4 worms were placed in the center of OP50 *E. coli* lawns on standard NGM plates and incubated at 20° C for 24 hours. Plates were examined 24 hours later and the numbers of worms remaining on the lawn or on unseeded agar were counted. Worms found on the sides of the dish or otherwise absent from NGM surface were scored as off the plate.

Brood size

For 24-hour brood assessments, young adult worms were placed on plates and allowed to lay eggs for 24 hours before being removed. Viable offspring were counted 48 hours later. For total brood assessments, L4 worms were placed on plates and moved to fresh plates every 24 hours for 5 days to ensure all eggs were accounted for. Viable offspring from eggs laid on each plate were counted 48 hours later and summed.

ZnMP fluorescence

L4 worms grown on OP50 plates were placed on plates treated with 40 μ M zinc mesoporphyrin (ZnMP; Santa Cruz) for 16 hours before being washed in M9 buffer, anesthetized in sodium axide, and mounted on slides with 2% agarose pads for imaging.

Worms in the starved condition were placed on unseeded plates for 16 hours before preparation and mounting as above. For liquid media experiments, three generations of worms were cultured in regular heme (20 μ M) axenic media, with the first two generations receiving antibiotic-supplemented media (10 mg/ml tetracycline) and the 3rd generation cultivated without antibiotic. L4 worms from the 3rd generation were placed in media containing 40 μ M ZnMP for 16 hours before being prepared and mounted for imaging as above. Worms were imaged on Zeiss Axio Imager 2 at 40x magnification, with image settings kept uniform across all images. Fluorescent intensity was measured within the proximal region of the intestine using ImageJ.

RNA-seq cluster and gene ontology analysis

The RNA sequencing (RNA-seq) dataset generated by Sun et al. was analyzed with Empirical Analysis of Gene Expression in R (edgeR) and the top 500 genes were extracted. Computational cluster analysis was done in RStudio. Gene ontology analysis was performed on significantly enriched genes using WormCat.²⁸ Source code is available in supplemental materials.

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
strain (Caenorhabditis elegans)	N2	Caenorhabditis Genetics Center (CGC) at the University of Minnesota	Bristol N2	
strain (Caenorhabditis elegans)	Double knockout (DKO)	Chen Lab, Zhejiang University	CCH303 (hrg-9(cck301)V; hrg-10(cck302)V)	Sun et al., 2022
strain (Caenorhabditis elegans)	eat-2	Samuelson Lab, University of Rochester	eat-2(ad465)	Raizen & Avery, 1995
chemical compound, reagent	gallium protoporphyrin IX chloride (GaPP)	Santa Cruz	CAS 210409-12-4	GaPP used in Sun et al., 2022 sourced from Frontier Scientific
chemical	Zn(II)	Santa	CAS 14354-67-	ZnMP used in Sun et al., 2022

Key resources table.

compound, reagent	Mesoporphyrin IX (ZnMP)	Cruz	7	sourced from Frontier Scientific
software, algorithm	Empirical Analysis of Gene Expression in R (edgeR)	Bioconductor.org		Chen et al., 2025
software, algorithm	WormCat	Github.com		Holdorf et al., 2020
chemical compound, reagent	Paraquat (methyl viologen dichloride hydrate)	Sigma Aldrich	856177	
chemical compound, reagent	TRIzol	Invitrogen	15596026	
commercial assay or kit	RNeasy Mini Kit	Qiagen	74104	
commercial assay or kit	Maxima First Strand cDNA Synthesis Kit	Thermo Fisher	K1671	
chemical compound, reagent	iTaq Universal SYBR Green Supermix	Invitrogen	1725121	
recombinant DNA reagent	<i>hrg-9</i> F	This paper	PCR primers	GGACCCGCTGCCATACACTA ATC
recombinant DNA reagent	<i>hrg-9</i> R	This paper	PCR primers	GACAATTCAAATCTGGCATCG TG
recombinant DNA reagent	<i>hrg-10</i> F	This paper	PCR primers	AGGCTTCCCGGAGCACATTT AC
recombinant DNA reagent	<i>hrg-10</i> R	This paper	PCR primers	CAGGCTCCATGCGTCTATCC AG
recombinant DNA reagent	<i>act</i> F	This paper	PCR primers	CAACACTGTTCTTTCCGGAG
recombinant DNA reagent	<i>act</i> R	This paper	PCR primers	CTTGATCTTCATGGTTGATGG G
gene (Danio rerio)	<i>tango2</i>	ENSMBL	ENSDARG00000056550	Zebrafish homolog of TANGO2
strain, strain background (Danio rerio, Sex NA) Sex is not	TUAB	Zebrafish International Resource Center	TU (ZL57) AB (ZL1)	Wild-type lines

Key resources table. (continued)

determined in Danio Rerio at the age group animals used in this study.				
genetic reagent	<i>sgRNA to tango2</i>	ThermoFisher Scientific	Kim et al., 2023	Guide RNA to create mutations in tango2 gene
sequence-based reagent	<i>tango2 F</i>	Kim et al., 2023	PCR primer	ATGGCTGAAAGAGCTGTGCT
sequence-based reagent	tango2 R	Kim et al., 2023	PCR Primer	ATGGCTGAAAGAGCTGTGCT
chemical compound, drug	Alexa Fluor 488-Phalloidin	ThermoFisher Scientific	A12379	1:40 dilution
chemical compound, drug	Methylcellulose	Sigma-Aldrich	M0387-500G	1% w/v dilution
software, algorithm	Prism	Graphpad		
Special Instrumentation	Zebrafish automated activity monitor	Zantiks	ZantiksMWP	
gene (Saccharomyces cerevisiae)	ygr127w	Saccharomyces Genome Database (SGD)	S000003359	Yeast homolog of human Tango2
strain, strain background: S.cerevisiae, haploid, MATa	BY4741(MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0)	EUROSCARF		
strain, strain background: S.cerevisiae, haploid, MATα	BY4742 (his3Δ1 leu2Δ0 lys2Δ0 ura3Δ0)	EUROSCARF		
strain, strain background: S.cerevisiae, haploid, MATa	W301(leu2-3,112 trp1-1 can1-100 ura3-1 ade2-1 his3-11,15)	Lab stock		
genetic reagent	YGR127W::KanMX in	Lab stock, Chen Lab,	YGR127W::KanMX	Sun et al., 2022

Key resources table. (continued)

(<i>S. cerevisiae</i>)	BY4741, BY4742	Zhejiang University		
genetic reagent (include species here) (<i>S. cerevisiae</i>)	YGR127W::KanMX in W303	This paper	YGR127W::KanMX	Deletion strain generated by PCR replacement. verified by locus-specific PCR.
genetic reagent	sgRNA: <i>ygr127w</i>	Integrated DNA Technologies (IDT)		Guide RNA sequence used for CRISPR-Cas9-mediated deletion of YGR127W; replaced with KanMX resistance cassette to generate <i>ygr127wΔ</i> strain.
recombinant DNA reagent	pRS313-YGR127W (HIS3, CEN6)	This paper		Yeast centromeric plasmid expressing YGR127W used for complementation
sequence-based reagent	<i>ygr127wgR</i> NA-F	This paper	sgRNA: <i>ygr127w</i>	GACTTTAATACGAAATCAAGATCCCG
sequence-based reagent	<i>ygr127wgR</i> NA-R	This paper	sgRNA: <i>ygr127w</i>	AAACCGGGATCTTGATTTCGTATTAA
sequence-based reagent	<i>ygr127 F</i>	This paper	PCR Primer	AGTCGGATCCTCAAGGTTCTTCTCTAGAACC
sequence-based reagent	<i>ygr127 R</i>	This paper	PCR Primer	AGTCGGATCCTGCTCTATTG GAGAACTTAACC
sequence-based reagent	<i>ygr127 F</i>	This paper	PCR Primer	TTGGCATCTGCCTAGCTTTTCG
sequence-based reagent	<i>ygr127 R</i>	This paper	PCR Primer	AGCGTCTACTGTGGTTACTG
commercial assay or kit	Q5 High-Fidelity DNA Polymerase	New England Biolabs (NEB)		Cloning of inserts into pRS313 backbones.
commercial assay or kit	T4 DNA Ligase	New England Biolabs (NEB)		Cloning of inserts into pRS313 backbones.
commercial assay or kit	CutSmart Buffer and Restriction Enzymes	New England Biolabs (NEB)		Cloning of inserts into pRS313 backbones.

Key resources table. (continued)

	(EcoRI, XhoI, BamHI)			
commercial assay or kit	Golden Gate Assembly Kit (Type IIS cloning)	New England Biolabs (NEB)		Used for one-pot modular assembly of plasmid constructs
commercial assay or kit	EZ Yeast Transformation Kit	Zymo Research	Used for high-efficiency yeast transformation during plasmid integration and complementation assays.	

Key resources table. (continued)

Stress conditions and RT-qPCR

N2 worms were subjected to the following stress conditions prior to RNA extraction: 1) fasting: worms were deprived of OP50 *E. coli* for 24 hours; 2) heat: worms were incubated for 4 hours at 34° C; 3) paraquat: worms were placed on standard NGM/OP50 plates treated with 25 mM paraquat (methyl viologen dichloride hydrate; Sigma Aldrich) for 24 hours. Whole worm RNA was extracted with TRIzol (Invitrogen) and purified using the RNeasy Mini Kit (Qiagen). cDNA generation was performed with the Maxima First Strand cDNA Synthesis Kit (Thermo Fisher). qPCR was performed in triplicate using iTaq Universal SYBR Green Supermix (Invitrogen) with a CFX Duet Realtime qPCR machine (BioRad). Gene expression was normalized to worm *act* and analyzed using the $\Delta\Delta C_q$ method. Primers used were as follows:

hrg-9: GGACCCGCTGCCATACACTAATC and GACAATTCAAATCTGGCATCGTG

hrg-10: AGGCTTCCCGGAGCACATTTAC and CAGGCTCCATGCGTCTATCCAG

act: CAACACTGTTCTTTCCGGAG and CTTGATCTTCATGGTTGATGGG

Yeast strain construction

YGR127w was knocked out by homologous recombination. A strain (BY4741) harboring a *ygr127wD* was used as a template and the knockout cassette was amplified by PCR using the following primers that anneal upstream and downstream of the locus:

TTGGCATCTGCCTAGCTTTTCG and AGCGTCTACTGTGGTTACTG

The PCR amplicon was then transformed into W303-1a cells and transformants were selected on YPD plates containing 200mg/ml G418. Integration at the correct locus was confirmed by PCR using the same primers, as above. To complement the *ygr127wD* in the Sun et al. strain, wild type *YGR127w* was amplified with 400 base pairs on either side of the gene and cloned into a *HIS3*-containing plasmid (pRS413). Transformants were selected on synthetic complete (SC) medium lacking histidine.

Yeast growth curves

Cells were grown to stationary phase at 25°C in either SC or SC-histidine medium. Cultures in the same medium were inoculated at an OD600 of ~0.01 in a 100ml volume in a 96-well plate. The OD600 was read every 15 minutes on a Sunrise Tecan microplate reader.

Zebrafish

All procedures involving zebrafish were approved by the Brigham and Women's Hospital Animal Care and Use Committee. Fish were bred and maintained using standard methods as described.²⁹ tango2bwh210 and tango2bwh211 zebrafish lines were created in our laboratory by the CRISPR-Cas9 approach as described previously.²² The tango2bwg210 allele has an insertion of seven bases (c.226_227ins7; p.Tyr76Leufs*25) and tango2bwg211 has a 26 base insertions in exon2 (c.226_227ins26; p.Tyr76Leufs*207) resulting in frameshift mutations and loss of protein function. Whole mount phalloidin staining and microscopy was performed as described previously.³⁰

Statistics and Reproducibility

All statistical analyses were performed in GraphPad Prism 9. Data presented are mean $\pm$ S.E.M. One-way analysis of variance (ANOVA) followed by Bonferroni's multiple comparisons was used to determine statistical significance ($p < 0.05$). Sample sizes were not pre-determined.

Data availability

All raw data from worm behavior, yeast, zebrafish, and RNA-seq studies are available as a supplement to this paper.

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Additional information

Contributions

SJM, SES, KSY, APW, and LW designed the *C. elegans* experiments. SES, KSY, OG, and MBB performed the *C. elegans* experiments and analyzed the data. LDO performed the RNA-seq analysis. MS designed the yeast experiments. AN performed the yeast experiments and analyzed the data. VAG and ESK designed the zebrafish experiments, and EK performed these experiments. VAG and ESK analyzed the zebrafish data. SJM and SES wrote the initial draft of the manuscript. All authors reviewed the manuscript and contributed to its final version.

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Additional files

[Supplemental Figure](#)

[Extended Data](#)

[R Code](#)

[RNA Seq Output](#)

References

- Kremer L. S., et al. (2016) Bi-allelic Truncating Mutations in TANGO2 Cause Infancy-Onset Recurrent Metabolic Crises with Encephalocardiomyopathy. *Am J Hum Genet* **98**:358-362
<https://doi.org/10.1016/j.ajhg.2015.12.009>

- 2 Lalani S. R., et al. (2016) Recurrent Muscle Weakness with Rhabdomyolysis, Metabolic Crises, and Cardiac Arrhythmia Due to Bi-allelic TANGO2 Mutations. *Am J Hum Genet* **98**:347-357 <https://doi.org/10.1016/j.ajhg.2015.12.008>
- 3 Powell A. R., Ames E. G., Knierbein E. N., Hannibal M. C., Mackenzie S. J (2021) Symptom Prevalence and Genotype-Phenotype Correlations in Patients With TANGO2-Related Metabolic Encephalopathy and Arrhythmias (TRMEA). *Pediatr Neurol* **119**:34-39 <https://doi.org/10.1016/j.pediatrneurol.2021.02.011>
- 4 Jennions E., et al. (2019) TANGO2 deficiency as a cause of neurodevelopmental delay with indirect effects on mitochondrial energy metabolism. *J Inherit Metab Dis* **42**:898-908 <https://doi.org/10.1002/jimd.12149>
- 5 Bard F., et al. (2006) Functional genomics reveals genes involved in protein secretion and Golgi organization. *Nature* **439**:604-607 <https://doi.org/10.1038/nature04377>
- 6 Milev M. P., et al. (2021) The phenotype associated with variants in TANGO2 may be explained by a dual role of the protein in ER-to-Golgi transport and at the mitochondria. *J Inherit Metab Dis* **44**:426-437 <https://doi.org/10.1002/jimd.12312>
- 7 Lujan A. L., et al. (2025) TANGO2 is an acyl-CoA binding protein. *J Cell Biol* **224** <https://doi.org/10.1083/jcb.202410001>
- 8 Heiman P., et al. (2022) Mitochondrial dysfunction associated with TANGO2 deficiency. *Sci Rep* **12**:3045 <https://doi.org/10.1038/s41598-022-07076-9>
- 9 Lujan A. L., et al. (2023) Defects in lipid homeostasis reflect the function of TANGO2 in phospholipid and neutral lipid metabolism. *eLife* **12** <https://doi.org/10.7554/eLife.85345>
- 10 Mehranfar M., et al. (2024) Lipidomic analysis of human TANGO2-deficient cells suggests a lipid imbalance as a cause of TANGO2 deficiency disease. *Biochem Biophys Res Commun* **717** <https://doi.org/10.1016/j.bbrc.2024.150047>
- 11 Asadi P., Milev M. P., Saint-Dic D., Gamberi C., Sacher M (2022) Vitamin B5, a coenzyme A precursor, rescues TANGO2 deficiency disease-associated defects in Drosophila and human cells. *J Inherit Metab Dis* <https://doi.org/10.1002/jimd.12579>
- 12 Miyake C. Y., et al. (2022) Natural History of TANGO2 Deficiency Disorder: Baseline Assessment of 73 Patients. *Genet Med* <https://doi.org/10.1016/j.gim.2022.11.020>
- 13 Sun F., et al. (2022) HRG-9 homologues regulate haem trafficking from haem-enriched compartments. *Nature* **610**:768-774 <https://doi.org/10.1038/s41586-022-05347-z>
- 14 Han S., Guo K., Wang W., Tao Y. J., Gao H (2023) Bacterial TANGO2 homologs are heme-trafficking proteins that facilitate biosynthesis of cytochromes c. *mBio* **14**:e0132023 <https://doi.org/10.1128/mbio.01320-23>
- 15 Jayaram D. T., et al. (2025) Heme allocation in eukaryotic cells relies on mitochondrial heme export through FLVCR1b to cytosolic GAPDH. *Nat Commun* <https://doi.org/10.1038/s41467-025-62819-2>
- 16 Lakowski B., Hekimi S (1998) The genetics of caloric restriction in *Caenorhabditis elegans*. *Proc Natl Acad Sci U S A* **95**:13091-13096 <https://doi.org/10.1073/pnas.95.22.13091>
- 17 Greer E. L., Brunet A (2009) Different dietary restriction regimens extend lifespan by both independent and overlapping genetic pathways in *C. elegans*. *Aging Cell* **8**:113-127 <https://doi.org/10.1111/j.1474-9726.2009.00459.x>
- 18 Van Raamsdonk J. M., et al. (2010) Decreased energy metabolism extends life span in *Caenorhabditis elegans* without reducing oxidative damage. *Genetics* **185**:559-571 <https://doi.org/10.1534/genetics.110.115378>
- 19 Byrne J. J., et al. (2019) Disruption of mitochondrial dynamics affects behaviour and lifespan in *Caenorhabditis elegans*. *Cell Mol Life Sci* **76**:1967-1985 <https://doi.org/10.1007/s00018-019-03024-5>
- 20 Ghosh R., Emmons S. W (2008) Episodic swimming behavior in the nematode *C. elegans*. *J Exp Biol* **211**:3703-3711 <https://doi.org/10.1242/jeb.023606>

- 21 Dimitrov L. N., Brem R. B., Kruglyak L., Gottschling D. E (2009) Polymorphisms in multiple genes contribute to the spontaneous mitochondrial genome instability of *Saccharomyces cerevisiae* S288C strains. *Genetics* **183**:365-383 <https://doi.org/10.1534/genetics.109.104497>
- 22 Kim E. S., et al. (2023) Intrinsic and extrinsic regulation of rhabdomyolysis susceptibility by Tango2. *Dis Model Mech* **16** <https://doi.org/10.1242/dmm.050092>
- 23 Yuan X., Protchenki O., Philpott C. C., Hamza I (2011) Topologically Conserved Residues Direct Heme Transport in HRG-1-related Proteins. *J Biol Chem* **287**:4914-4924 <https://doi.org/10.1074/jbc.M111.326785>
- 24 Chen C., et al. (2011) An intercellular heme trafficking protein delivers maternal heme to the embryo during development in *C. elegans*. *Cell* **145**:720-732 <https://doi.org/10.1016/j.cell.2011.04.025>
- 25 Homan R. A., Jadhav A. M., Conway L. P., Parker C. G (2022) A Chemical Proteomic Map of Heme-Protein Interactions. *J Am Chem Soc* **144**:15013-15019 <https://doi.org/10.1021/jacs.2c06104>
- 26 Soo S. K., Van Raamsdonk J. M (2021) High confidence ATFS-1 target genes for quantifying activation of the mitochondrial unfolded protein response. *MicroPubl Biol* 2021 <https://doi.org/10.17912/micropub.biology.000484>
- 27 Di Pede A., et al. (2025) Mild activation of the mitochondrial unfolded protein response increases lifespan without increasing resistance to stress. *Open Biol* **15** <https://doi.org/10.1098/rsob.240358>
- 28 Holdorf A. D., et al. (2020) WormCat: An Online Tool for Annotation and Visualization of *Caenorhabditis elegans* Genome-Scale Data. *Genetics* **214**:279-294 <https://doi.org/10.1534/genetics.119.302919>
29. Westerfield M (2000) *The Zebrafish Book: A Guide for the Laboratory Use of Zebrafish (Danio Rerio)* Institute of Neuroscience. University of Oregon.
- 30 Casey J. G., et al. (2023) NRAP reduction rescues sarcomere defects in nebulin-related nemaline myopathy. *Hum Mol Genet* **32**:1711-1721 <https://doi.org/10.1093/hmg/ddad011>

Peer reviews

Reviewer #1 (Public review):

Sandkuhler et al. re-evaluated the biological functions of TANGO2 homologs in *C. elegans*, yeast, and zebrafish. Compared to the previously reported role of TANGO2 homologs in transporting heme, Sandkuhler et al. expressed a different opinion on the biological functions of TANGO2 homologs. With the support of some results from their tests, they conclude that 'there is insufficient evidence to support heme transport as the primary function of TANGO2', in addition to the evidence that *C. elegans* TANGO2 helps counteract oxidative stress.. While the differences are reported in this study, more work is needed to elucidate the intuitive biological function of TANGO2.

Strengths:

(1) This work revisits a set of key experiments, including the toxic heme analog GaPP survival assay, the fluorescent ZnMP accumulation assay, and the multi-organismal investigations documented by Sun et al. in *Nature* (2022), which are critical for comparing the two works. Meanwhile, the authors also highlight the differences in reagents and methods between the two studies, demonstrating significant academic merit.

(2) This work reported additional phenotypes for the *C. elegans* mutant of the TANGO2 homologs, including lawn avoidance, reduced pharyngeal pumping, smaller brood size, faster exhaustion under swimming test, and a shorter lifespan. These phenotypes are important for understanding the biological function of TANGO2 homologs, while they were missing from the report by Sun et al.

(3) Investigating the 'reduced GaPP consumption' as a cause of increased resistance against the toxic GaPP for the TANGO2 homologs, hrg-9 hrg-10 double null mutant provides a valuable perspective for studying the biological function of TANGO2 homologs.

(4) The induction of hrg-9 gene expression by paraquat indicates a strong link between TANGO2 and mitochondrial function.

(5) This work thoroughly evaluated the role of TANGO2 homologs in supporting yeast growth using multiple yeast strains and also pointed out the mitochondrial genome instability feature of the yeast strain used by Sun et al.

Weakness:

It is always a challenge to replicate someone else's work, but it is worthwhile to take on the challenge, provide evidence, and raise concerns about it. These authors attempted to replicate the experiment using the same biological material as that used by Sun et al. in Nature (2022), despite some experimental differences between the two studies. This study does not have many technical weaknesses, but it can become a much better project by focusing on the new phenotypes discovered here.

<https://doi.org/10.7554/eLife.105418.2.sa3>

Reviewer #2 (Public review):

This work offers a valuable re-evaluation of earlier claims from other groups about TANGO2 functions and proposes that energy-related and stress-related pathways may be more important to the disorder than previously thought. A key strength of this work is the use of multiple model systems. The authors provide solid data that show how TANGO2 is probably only indirectly involved in heme transport and provide support for alternative mechanisms where TANGO2 is actually directly control. These findings provide valuable information for researchers seeking more accurate therapeutic targets.

Strengths:

The study refutes earlier claims about TANGO2's involvement in heme transport and extends previous findings by implicating TANGO2 in metabolism and oxidative stress, thereby highlighting new aspects of its role in cell physiology. The use of different model systems (*Saccharomyces cerevisiae*, *Caenorhabditis elegans*, *Danio rerio*) to address the main research questions is useful and demonstrates evolutionary conservation of the studied processes. Finally, the results suggest a broader impact than previously described, somewhat supporting the novelty of the study.

Weaknesses:

Although the phenotypic analyses are broad and generally well executed, a key limitation is that the main conclusions mainly rely on these readouts. While informative, sole phenotypic analyses cannot directly demonstrate the underlying molecular mechanisms proposed by the authors. The study includes limited functional or biochemical assays connecting TANGO2 orthologs to the proposed energy and stress pathways. Some observations would benefit from additional orthogonal validation to strengthen the overall interpretation. As a result, the evidence supporting the central mechanistic interpretation remains indirect, although compelling.

Overall, the authors have achieved their stated aims, and their results mainly support their main conclusion (i.e., TANGO2 is unlikely to function in heme transport and is probably linked to energy and stress pathways). However, much of the evidence comes from

phenotypic analyses, which limits the strength of the mechanistic claims, leaving the proposed pathways somewhat indirect.

This work is likely to have a valuable impact on the subfield by clarifying that TANGO2 is not involved (at least directly) in heme transport and clarifying its actual role in energy and stress-related processes. By rigorously reassessing and confuting earlier claims from other studies across multiple model systems, the current work will help to guide the future research and therapeutic exploration in the context of TANGO2 deficiencies. This study will provide a solid foundation for more mechanistic insights into TANGO2 function.

<https://doi.org/10.7554/eLife.105418.2.sa2>

Reviewer #3 (Public review):

In this paper, Sandkuhler et al. reassessed the role of TANGO2 as a heme chaperone proposed by Sun et al in a recently published paper (<https://doi.org/10.1038/s41586-022-05347-z>). Overall, Sandkuhler et al. conclude that the heme-related roles of TANGO2 had been overemphasized by Sun et al. especially because the *hrg9* gene does not exclusively respond to different regimens of heme synthesis/uptake but is susceptible to a greater extent to, for example, oxidative stress. Impaired heme trafficking is then interpreted as due to general mitochondrial dysfunction. In recent years, the discussion around the heme-related roles of TANGO2 has been tantalizing but is still far from a definitive consensus. Discrepancies between results and their interpretation are testament to how ambitious the understanding of TANGO2 and the phenotypes associated with TANGO2 defects are.

The work presented by Sandkuhler et al. is methodologically sound, and the authors have appropriately addressed my concerns in the first round of review. Overall, this paper challenges the recent developments in the field in relation to heme trafficking and provides a wider perspective on the biological roles of TANGO2.

<https://doi.org/10.7554/eLife.105418.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) A detailed comparison between this work and the work of Sun et al. on experimental protocols and reagents in the main text will be beneficial for readers to assess critically.

We have added a Key Reagents Table outlining the key reagents used in our study. In terms of experimental protocols, we replicated those described by Sun et al. in most instances and described any differences when present. With this resubmission, we included additional ZnMP accumulation experiments in liquid media (see point 3 below).

(2) The GaPP used by Sun et al. (purchased from Frontier Scientific) is more effective in killing the worm than the one used in this study (purchased from Santa Cruz). Is the different outcome due to the differences in reagents? Moreover, Sun et al. examined the lethality after 3-4 days, while this work examined the lethality after 72 hours. Would the extra 24 hours make any difference in the result?

We now cite product vender differences as a possible reason for the observed difference in worm death, as the reviewer suggests, on page 8 (see text below) and include these differences in the Key Reagents Table. We also now stress the fact that our experiments included different doses of GaPP and the use of *eat-2* mutants as an additional control, which we believe adds rigor and demonstrates the potency of GaPP in our experiments. We decided

on assessment at 72 hours, as we deemed it a less nebulous time point as compared to 3-4 days. Most of the observed worm death occurred earlier in this interval, so we believe it is unlikely that large group differences would emerge after an additional 24 hours.

“Exposing worms to GaPP, a toxic heme analog, we observed that nematodes deficient in HRG-9 and HRG-10 displayed increased survival compared to WT worms, consistent with prior work,[13] though the between-group difference was markedly smaller in our study. We required higher GaPP concentrations to induce lethality, potentially due to product vendor differences, but did observe a clear dose-dependent effect across strains. Although it was previously proposed that the survival benefit seen in worms lacking HRG-9 and HRG-10 resulted from reduced transfer from intestinal cells after GaPP ingestion, our data suggest the reduced lethality is more likely due to decreased environmental GaPP uptake. Supporting this notion, DKO worms exhibited lawn avoidance, reduced pharyngeal pumping, and modestly lower intestinal ZnMP accumulation when exposed to this fluorescent heme analog on agar plates. In liquid media, DKO worms demonstrated higher fluorescence, but only in ZnMP-free conditions, suggesting the presence of gut granule autofluorescence. Furthermore, survival following exposure to GaPP was highest in eat-2 mutants, despite heme trafficking being unaffected in this strain.”

(3) This work reported the opposite result of Sun et al. for the fluorescent ZnMP accumulation assay. However, the experimental protocols used by the two studies are massively different. Sun et al. did the ZnMP staining by incubating the L4-stage worms in an axenic mCeHR2 medium containing 40 μ M ZnMP (purchased from Frontier Scientific) and 4 μ M heme at 20 °C for 16 h, while this work placed the L4-stage worms on the OP50 E. coli seeded NGM plates treated with 40 μ M ZnMP (purchased from Santa Cruz) for 16 h. The liquid axenic mCeHR2 medium is bacteria-free, heme-free, and consistent for ZnMP uptake by worms. This work has mentioned that the hrg-9 hrg-10 double null mutant has bacterial lawn avoidance and reduced pharyngeal pumping phenotypes. Therefore, the ZnMP staining protocol used in this work faces challenges in the environmental control for the wild type vs. the mutant. The authors should adopt the ZnMP staining protocol used by Sun et al. for a proper evaluation of fluorescent ZnMP accumulation.

We agree with this comment. As such, we performed the ZnMP assay in liquid media conditions, as now described on page 13:

“For liquid media experiments, three generations of worms were cultured in regular heme (20 μ M) axenic media, with the first two generations receiving antibiotic-supplemented media (10 mg/ml tetracycline) and the 3rd generation cultivated without antibiotic. L4 worms from the 3rd generation were placed in media containing 40 μ M ZnMP for 16 hours before being prepared and mounted for imaging as above. Worms were imaged on Zeiss Axio Imager 2 at 40x magnification, with image settings kept uniform across all images. Fluorescent intensity was measured within the proximal region of the intestine using ImageJ.”

In heme-free media, both WT and DKO worms invariably entered L1 arrest, thus we were not able to replicate the results reported by Sun et al. Using media containing heme, we did see an increase in fluorescence, but this was only in the ZnMP-free condition, indicating that the increased signal was attributable to autofluorescence. This is a known phenomenon associated with gut granules in C. elegans in the setting of oxidative stress. The results of these experiments are now summarized on page 6:

“DKO nematodes at the L4 larval stage were previously shown to accumulate the fluorescent heme analog zinc mesoporphyrin IX (ZnMP) in intestinal cells in low-heme (4 μ M) liquid media. While attempting to replicate this experiment, we observed that both wildtype and DKO nematodes entered L1 arrest under these conditions. Therefore, to allow for

developmental progression, we grew worms on standard OP50 *E. coli* plates and in media containing physiological levels of heme (20 μ M). We then examined whether differences in ZnMP uptake persisted under these basal conditions. DKO worms grown on ZnMP-treated *E. coli* plates displayed significantly reduced intestinal ZnMP fluorescence compared to N2 (Figure 1B and C). Using basal heme media with ZnMP, there was no significant difference in ZnMP fluorescence between DKO and wildtype nematodes, although DKO worms grown in media without ZnMP exhibited significantly higher autofluorescence (Figure 1D and E). To test whether autofluorescence may have contributed to the higher fluorescent intensities previously reported in heme-deficient DKO worms, we repeated this experiment on agar plates under starved conditions but did not observe a difference between groups (Figure 1B)."

(4) A striking difference between the two studies is that Sun et al. emphasize the biochemical function of TANGO2 homologs in heme transporting with evidence from some biochemical tests. In contrast, this work emphasizes the physiological function of TANGO2 homologs with evidence from multiple phenotypical observations. In the discussion part, the authors should address whether these observed phenotypes in this study can be due to the loss of heme transporting activities upon eliminating TANGO2 homologs. This action can improve the merit of academic debate and collaboration.

Thank you for this suggestion. The following text has been added to the Discussion section (page 9):

"In addition to altered pharyngeal pumping, DKO worms displayed multiple previously unreported phenotypic features, suggesting a broader metabolic impairment and reminiscent of some clinical manifestations observed in patients with TDD. Elucidating the mechanisms underlying this phenotype, and whether they reflect a core bioenergetic defect, is an active area of investigation in our lab. Several *C. elegans* heme-responsive genes have been characterized, revealing relatively specific defects in heme uptake or utilization rather than broad organismal dysfunction. For example, *hrg-1* and *hrg-4* mutants exhibit impaired growth only under heme-limited conditions,[23] and *hrg-3* loss affects brood size and embryonic viability specifically when maternal heme is scarce.[24]]By contrast, *hrg-9* and *hrg-10* mutants exhibit the most severe organismal phenotypes of the *hrg* family, to date, including reduced pharyngeal pumping, decreased motility, shortened lifespan, and smaller broods, even when fed a heme-replete diet."

Reviewer #2 (Public review):

(1) The manuscript is written mainly as a criticism of a previously published paper. Although reproducibility in science is an issue that needs to be acknowledged, a manuscript should focus on the new data and the experiments that can better prove and strengthen the new claims.

Thank you for this suggestion. While the primary intent of this study was to replicate key findings from the 2022 publication by Sun et al., the revised manuscript now emphasizes underlying mechanisms more broadly rather than focusing narrowly on that prior publication.

(2) The current presentation of the logic of the study and its results does not help the authors deliver their message, although they possess great potential.

We have attempted to rectify this through substantial revision of the Discussion section and other places throughout the manuscript.

*(3) The study is missing experiments to link *hrg-9* and *hrg-10* more directly to bioenergetic and oxidative stress pathways.*

The reviewer is correct in this assertion, but it was not our intent to definitively prove this link or, indeed, the primary mechanism of TANGO2 in the present manuscript. This said, we are actively engaged in this endeavor in our lab and anticipate these data will be published in a separate, forthcoming publication.

We have added additional references pertaining to hrg-9 enrichment as part of the mitochondrial unfolded protein response (page 10) and a comparison of the phenotype observed in hrg-9 and hrg-10 deficient worms versus those lacking other proteins in the hrg family (page 9).

Reviewer #3 (Public review):

(1) The authors stress - with evidence provided in this paper or indicated in the literature - that the primary role of TANGO2 and its homologues is unlikely to be related to heme trafficking, arguing that observed effects on heme transport are instead downstream consequences of aberrant cellular metabolism. But in light of a mounting body of evidence (referenced by the authors) connecting more or less directly TANGO2 to heme trafficking and mobilization, it is recommended that the authors comment on how they think TANGO2 could relate to and be essential for heme trafficking, albeit in a secondary, moonlighting capacity. This would highlight a seemingly common theme in emerging key players in intracellular heme trafficking, as it appears to be the case for GAPDH - with accumulating evidence of this glycolytic enzyme being critical for heme delivery to several downstream proteins.

TANGO2 is essential for mitochondrial health, albeit in a yet unknown capacity. In the absence of TANGO2, defects in heme trafficking may be secondary sequelae of mitochondrial dysfunction. We would point out that prior studies that attempted to show that TANGO2 and its homologs are involved in heme trafficking proposed very different mechanisms (direct binding vs. membrane protein interaction) and relied on artificially low or high heme conditions to produce these effects. We have attempted to address these more clearly in the Discussion section and have added a fifth figure to summarize our current unifying theory for how heme levels and mitochondrial stress may be linked.

(2) The observation - using eat-2 mutants and lawn avoidance behaviour - that survival patterns can be partially explained by reduced consumption, is fascinating. It would be interesting to quantify the two relative contributions.

We have completed additional ZnMP experiments in liquid media at the reviewers' request. This experimental condition eliminates lawn avoidance as a factor in consumption. Fluorescent intensity was significantly higher in the DKO worms in media lacking ZnMP, indicating increased autofluorescence in DKO worms, while signal was not significantly different in media with ZnMP.

(3) In the legend to Figure 1A it's a bit unclear what the differently coloured dots represent for each condition. Repeated measurements, worms, independent experiments? The authors should clarify this.

The following sentence has been added to the legend for Figure 1:

“Each dot represents the number of offspring laid by one adult worm on one GaPP-treated plate after 24 hours.”

(4) It would help if the entire fluorescence images (raw and processed) for the ZnMP treatments were provided. Fluorescence images would also benefit Figure 1B.

Fluorescent intensity values pertaining to the ZnMP experiments are included in our Extended Data supplement, and we have added representative images to Figure 1, per the

reviewer's request. We thank the reviewer for this helpful suggestion. We would be happy to upload raw images to an open-access repository if deemed necessary by the editorial team.

(5) Increasingly, the understanding of heme-dependent roles relies on transient or indirect binding to unsuspected partners, not necessarily relying on a tight affinity and outdated the notion of heme as a static cofactor. Despite impressive recent advancements in the detection of these interactions (for example <https://doi.org/10.1021/jacs.2c06104>; cited by the authors), a full characterisation of the hemome is still elusive. Sandkuhler et al. deemed it possible but seem to question that heme binding to TANGO2 occurs. However, Sun et al. convincingly showed and characterised TANGO2 binding to heme. It is recommended that the authors comment on this.

We believe it is plausible that TANGO2 binds heme (as do hundreds of other proteins), especially as it has been shown to bind other hydrophobic molecules. However, we also note that a separate paper examining the role of TANGO2 in heme transport posited that GAPDH is the sole heme binding partner for cytoplasmic transport (<https://doi.org/10.1038/s41467-025-62819-2>), contradicting the originally posited theory of how TANGO2 functions. This is described in the Discussion section and, as noted above, we have added an additional figure to demonstrate our unifying hypothesis for why TANGO2 may be important in the low-heme state, irrespective of any direct effect on heme trafficking.

Additional comments and revisions:

(1) It was suggested that a triple mutant (eat-2; hrg-9; hrg-10) be tested to determine the primary driver of GaPP toxicity. We appreciate this suggestion, but we offer the following rationale for why these experiments were not pursued. The eat-2 mutant, which lacks a nicotinic acetylcholine receptor subunit in pharyngeal muscles, was included solely as a dietary restriction control to illustrate that reduced GaPP toxicity in the hrg-9/10 double mutant could arise from poor feeding rather than defective heme transport. Both eat-2 and hrg-9/10 mutants exhibit markedly reduced feeding but via different mechanisms. In our assays, GaPP survival was inversely correlated with ingestion rate: eat-2 animals, which feed the least, showed the highest survival, while hrg-9/10 mutants showed intermediate feeding and intermediate survival. Consistent with this, eat-2 worms also displayed the lowest ZnMP accumulation.

(2) GaPP solution was added to NGM plates after seeding with OP50. This is now expressly stated in the Methods section (page 15). We would note that Sun et al. mixed GaPP in with NGM in the liquid phase. We would expect that if there were a difference in GaPP exposure due to these different protocols, worms in our experiment would have received higher GaPP concentrations.

“Standard NGM plates were treated with 1, 2, 5, or 10 μ M gallium protoporphyrin IX (GaPP; Santa Cruz) after seeding with OP50. Plates were swirled to ensure an even distribution of GaPP and allowed to dry completely.

(3) The manuscript has been reworked to read as more of an independent study rather than a rebuttal of prior work, though the primary objective of validating prior work remains unchanged.

(4) Several technical details of experiments have been moved from the main text to the materials and methods section.

(5) One reviewer noted that the figure numbering should be adjusted. Numbering does not progress sequentially (i.e., 1A...1B...2A...2B) early in the text, because we have opted to consolidate data pertaining to heme analog experiments in Figure 1 and behavioral data in Figure 2.

(6) “Kingdoms” has been changed to “domains” (page 4).

(7) Example images are now included for Figure 1B, as noted above.

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